

Lecture 9 Notes: Interest Rates and Yield Curves

Corporate Finance Chapter 6

I. Opening Remarks and Exam Philosophy

Perspective on Exams and Challenges

- Do not let small inconveniences (like light or fans) affect your focus during exams.
- In a session system, the paper can come from anywhere in the content taught.
- Topics stressed and repeated in class are the ones that will definitely appear on the exam.
- Professional life requires resilience—small issues should not be overvalued or cause undue stress.

II. Professional Qualifications: CFA and FRM

Finance students should consider professional examinations that complement their degree:

Qualification	Focus
CFA (Chartered Financial Analyst)	Investment analysis, portfolio management, stock analysis
FRM (Financial Risk Manager)	Financial risk management, derivatives, market risk
CA (Chartered Accountant)	Accounting, auditing, taxation

Exam Requirements

- Success depends on two key factors: **conceptualization** and **memory**.
- **Example:** 120 questions in 180 minutes = 1.5 minutes per question.
- Formulas must be memorized; there is no time to derive them during the exam.
- This pressure tests the ability to absorb information and perform under constraints—skills essential in professional careers.

Career Path

- After completing a Bachelor's degree, moving toward the CFA is highly recommended.
- The qualification is financially rewarding and globally recognized.

III. Determinants of Interest Rates

Real vs. Nominal Rates

- **Real Risk-Free Rate (r^*):** The starting point for required return on any debt security. Does not include an inflation premium.
- **Nominal Rate:** The quoted rate that includes inflation expectations.
- **Real Return:** What an investor actually earns after accounting for inflation.

$$\text{Nominal Rate} = r^* + IP + DRP + LP + MRP$$

Component Breakdown

Component	Description
r^*	Real Risk-Free Rate
IP	Inflation Premium: Compensates for expected inflation
DRP	Default Risk Premium: Risk of issuer default (zero for Treasuries)
LP	Liquidity Premium: Compensation for illiquidity
MRP	Maturity Risk Premium: Price sensitivity for longer maturities

Key Relationships

- **Treasury bonds:** No default risk premium ($DRP = 0$).
- **Corporate bonds:** Include DRP and LP based on credit quality.
- **Interest Rate Risk:** Prices and interest rates have an **inverse relationship**. As rates increase, bond prices decrease.
- **Reinvestment Risk:** Short-term securities face the risk that maturing funds must be reinvested at lower rates if market rates fall.

IV. Practice Problem: Interest Rate Calculation

Problem Statement

An analyst has the following information:

- Real Rate (r^*): 2% (constant for 3 years)
- Expected Inflation: Year 1 = 3%; Year 2 = 3.5%; Year 3 = 4%
- MRP Formula: $0.1 \times (t - 1)\%$ (where t = years to maturity)
- 3-Year Corporate Bond: $LP = 0.25\%$; $DRP = 0.6\%$

Calculate:

1. Rate for a 1-year T-bill
2. Rate for a 3-year T-bond
3. Rate for a 3-year Corporate bond

Step 1: Calculate Inflation Premiums (IP)

The inflation premium for a t -year security is the average of expected inflation over the t years.

$$IP_1 = 3.00\%$$
$$IP_3 = \frac{3\% + 3.5\% + 4\%}{3} = \frac{10.5\%}{3} = 3.50\%$$

Step 2: Calculate Maturity Risk Premiums (MRP)

$$MRP = 0.1 \times (t - 1)\%$$

$$MRP_1 = 0.1 \times (1 - 1)\% = 0.1 \times 0\% = 0.00\%$$

$$MRP_3 = 0.1 \times (3 - 1)\% = 0.1 \times 2\% = 0.20\%$$

Step 3: Calculate 1-Year T-bill Rate

For T-bills: $DRP = 0$, $LP = 0$

$$\begin{aligned} r_{T-bill} &= r^* + IP_1 + MRP_1 + DRP + LP \\ &= 2\% + 3.00\% + 0.00\% + 0\% + 0\% \\ &= \mathbf{5.00\%} \end{aligned}$$

Step 4: Calculate 3-Year T-bond Rate

For T-bonds: $DRP = 0$, $LP = 0$

$$\begin{aligned}r_{T-bond} &= r^* + IP_3 + MRP_3 + DRP + LP \\&= 2\% + 3.50\% + 0.20\% + 0\% + 0\% \\&= \mathbf{5.70\%}\end{aligned}$$

Step 5: Calculate 3-Year Corporate Bond Rate

For corporate bonds: $DRP = 0.6\%$, $LP = 0.25\%$

$$\begin{aligned}r_{Corp} &= r^* + IP_3 + MRP_3 + DRP + LP \\&= 2\% + 3.50\% + 0.20\% + 0.6\% + 0.25\% \\&= \mathbf{6.55\%}\end{aligned}$$

Final Answer

Results	
Security	Interest Rate
1-Year T-bill	5.00%
3-Year T-bond	5.70%
3-Year Corporate Bond	6.55%

V. Term Structure and Yield Curves

The relationship between different maturities and their interest rates is called the **Term Structure of Interest Rates**. When plotted graphically, it creates a **Yield Curve**.

Types of Yield Curves

Yield Curve Shape	Interpretation
Upward Sloping	Short-term rates $\uparrow$ Long-term rates. Occurs when MRP is positive and inflation is expected to rise.
Inverted (Downward Sloping)	Short-term rates $\downarrow$ Long-term rates. Occurs when inflation is expected to decrease significantly.
Humped	Intermediate-term rates $\downarrow$ both short-term and long-term rates. Implies inflation will rise temporarily then fall.

Flight to Quality

- During extreme uncertainty or war, investors shift funds to safe assets (Treasuries).
- This movement causes rapid shifts in interest rates across maturities.

- The risk premium between corporate and government bonds widens significantly.

VI. Summary of Key Concepts

1. **Interest Rate Components:** $r = r^* + IP + DRP + LP + MRP$
2. **Professional Qualifications:** CFA and FRM are highly relevant for finance careers.
3. **Yield Curves:** Upward sloping (normal), inverted, or humped based on inflation expectations.
4. **Risk Types:** Interest rate risk (price sensitivity) and reinvestment risk (reinvestment at lower rates).
5. **Treasury vs. Corporate:** Treasuries have no DRP ; corporate bonds include DRP and LP .